

On Equivalent Sets of Elements in a Free Group

Author(s): J. H. C. Whitehead

Source: *Annals of Mathematics*, Vol. 37, No. 4 (Oct., 1936), pp. 782-800

Published by: Mathematics Department, Princeton University

Stable URL: <https://www.jstor.org/stable/1968618>

Accessed: 03-04-2026 18:04 UTC

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



Mathematics Department, Princeton University is collaborating with JSTOR to digitize, preserve and extend access to *Annals of Mathematics*

ON EQUIVALENT SETS OF ELEMENTS IN A FREE GROUP

By J. H. C. WHITEHEAD

(Received January 2, 1936; revised March 26, 1936)

1. Let G be the free group generated by the symbols $a_1, \dots, a_p$ and their inverses¹ $a'_1, \dots, a'_p$, and let $(\alpha) = (\alpha_1, \dots, \alpha_m)$ be a set of elements of G which generate a sub-group Γ . In this paper we consider two kinds of equivalence between sets (α) and (β) . The first is equivalence under the automorphisms of Γ , generated by 'simple automorphisms' of the form

$$(1.1) \quad \alpha_\lambda^* = \alpha_\lambda \alpha_\rho^\epsilon \text{ or } \alpha_\rho^\epsilon \alpha_\lambda, \quad \alpha_\mu^* = \alpha_\mu \quad (\rho \neq \lambda, \quad \mu \neq \lambda, \quad \epsilon = \pm 1),$$

together with the 'simple automorphism' which replaces α_1 by its inverse. Relative to this kind of equivalence we have very little to add to a paper by J. Nielsen,² in which he gives a mechanical process for deciding whether or no two sets are equivalent and also a process for reducing (α) to one of a finite number of normal forms. When reduced in this way we shall describe a set of elements as *reduced* (N), and we recall that (α) is reduced (N) if it contains no two words of the form $(AB)^\epsilon$ and $(AC)^\epsilon$ respectively,³ where $l(A) > l(B)$ or $> l(C)$, and if the last half of every word with an even number of letters is an 'isolated ending.' That is to say, if $\alpha \equiv AB$ and $l(A) = l(B)$ no other word in (α) ends with B or begins with B' .

In §2 it is assumed that any empty words which appear⁴ during the process of reduction are discarded, as in J. N., while in §4 they are retained.

Theorem 1 below is essentially a restatement of various arguments used by Nielsen, while Theorem 2 adds a detail to J. N.

The second kind of equivalence refers to the effect on (α) of automorphisms⁵ of G , two ordered sets of elements (α) and (β) , both of which contain the same number of words, being equivalent if α_λ corresponds to β_λ ($\lambda = 1, 2, \dots$) in some automorphism of G . That is to say they are equivalent if there is an automorphism

$$a_i \rightarrow f_i(a, a') = f_i(a_1, \dots, a_p, a'_1, \dots, a'_p),$$

¹ As in a recent paper (J. H. C. Whitehead, *Proc. L. M. S.*, 41 (1936), 48-56, which will be referred to as W. G., W' will be used as an abbreviation for W^{-1} .

² *Mat. Tidsskrift*, (1921, B), 77-94. This paper will be referred to as J. N.

³ $l(W)$ stands for the number of letters in W . Here we assume that each word in (α) is reduced (E) (see the conclusion of this section).

⁴ This can happen after a transformation of the form (1.1), where $\alpha_\rho^\epsilon = \alpha_\lambda'$ but $\rho \neq \lambda$.

⁵ For a far reaching account of the group of automorphisms of G see Nielsen, *Math. Annalen*, 91 (1924), 169-209.

such that

$$\beta_\lambda(a, a') = \alpha_\lambda[f(a, a'), f'(a, a')].$$

Nielsen⁶ has shown that the automorphisms of G are generated by simple automorphisms of the form $a_i \rightarrow a_i a_j^{\pm 1}$, $a_k \rightarrow a_k$ ($j, k \neq i$), together with $a_1 \rightarrow a_1'$. Thus the problem is to determine whether or no an ordered set (α) can be transformed into an ordered set (β) by a sequence of what I have called *simple transformations*,⁷ among which we now include $a_1 \rightarrow a_1'$. In §3 this problem is solved for finite sets. We adopt the convention followed in W. G. relative to true and cyclic words, a cyclic word being a sequence of letters in cyclic order, and in §3 any statement about a set of words will apply equally well to a set of true and to a set of cyclic words. Thus the equivalence problem is solved both for ordered sets of elements and for ordered sets of classes of elements.

We shall follow H. Seifert and W. Threlfall⁸ in using the symbol $\equiv$ to denote equality between words and $=$ for equality between elements. Thus the true word $abb' = a$ but $\neq a$ and the cyclic word $bab' = a$. If $W(a, a')$ contains no syllable xx' we shall describe it as 'reduced (E)' or as 'reduced (E, a)', and 'reduced (E, α)' will have a similar meaning relative to $F(\alpha, \alpha')$, where $\alpha_\lambda = \alpha_\lambda(a, a')$.

2. Let Γ be the sub-group of G generated by (α) , where

$$\alpha_\lambda = \alpha_\lambda(a, a'),$$

and α_λ is reduced (E). Let

$$\begin{aligned} W(a, a') &\equiv F(\alpha, \alpha') \\ &= W^0(a, a'), \end{aligned}$$

be any element in Γ , where F is reduced (E, α) and W^0 is reduced (E, a). We represent W by an oriented polygonal segment $S(W)$, in a Euclidean plane, the oriented edges of which represent the letters of W in their correct order, a_i pointing the same way as $S(W)$ and a_i' pointing backwards. Whenever a syllable xx' is omitted during the reduction $W \rightarrow W^0$ we identify the corresponding edges of $S(W)$ in such a way that their common end point is on the left hand side of the new $S(W)$, positively described. Finally $S(W)$ will have been transformed into a figure $S^0(W)$, consisting of a segment $S^0(W^0)$ with a row of trees $T_1, \dots, T_q$ attached to its left hand side, and in spelling out W each edge of $S^0(W^0)$ is described once and each edge of T_p twice, with opposite senses. The

⁶ *Math. Annalen*, 79 (1919), 269-72.

⁷ W. G., §1. A simple transformation of a set of words (W), consists either of replacing x^e by $(xy)^e$ ($y \neq x$ or x') whenever x^e occurs in (W) , or of an elementary transformation, which consists of omitting a syllable xx' (elementary reduction) or of inserting one (elementary insertion).

⁸ *Lehrbuch der Topologie*, Leipzig (1934), chap. 12.

vertex in which T_p meets $S^0(W^0)$ will be called its *base*, and the other extremities of T_p will be called its *leaves*. It is to be understood that each tree has but a single edge incident with its base. Thus two or more trees may have a common base. By the length $l(s)$, of any segment⁹ s in $S^0(W)$ we shall mean the number of edges in s , and h_p will stand for the longest segment joining a leaf of T_p to its base. We assume $T_1, \dots, T_q$ to be the order in which the trees occur in spelling out W and shall denote by l_p the length of the segment, possibly zero, joining the base of T_p to the base of T_{p+1} , T_0 and T_{q+1} being the first and last vertices of $S^0(W^0)$. Finally, we shall describe T_p as a *stump* if it has but a single leaf.

THEOREM 1. *If (α) is reduced (N) , then each tree in $S^0(W)$ is a stump, no word in $F(\alpha, \alpha')$ ends at a vertex of T_p other than a leaf, and*

$$(2.1) \quad \begin{cases} a \left\{ \begin{array}{l} l_p \geq |h_{p+1} - h_p| \quad (h_0 = h_{q+1} = 0), \\ l_p > 0. \end{array} \right. \end{cases}$$

This theorem, being vacuous if $F \equiv 1$, will be proved by induction on $l(F)$, measured in terms of the α 's. We first show that no word in F , except possibly the last, ends at a vertex P , of T_p , which is not a leaf. For if it does, let

$$F(\alpha, \alpha') \equiv F_1(\alpha, \alpha')F_2(\alpha, \alpha'),$$

where F_1 ends at P . If P is the base of T_p let L be any leaf, and if P is not the base let L be any leaf which is separated from the base by P . Then the branch consisting of all the edges of T_p which are not separated from L by P is a tree belonging either to F_1 or to F_2 , say to F_1 . If

$$W_1(a, a') = F_1(\alpha, \alpha'),$$

the figure $S^0(W_1)$ has a tree growing out of its last vertex, contradicting the hypothesis $l_q > h_q$, calculated for $S^0(W_1)$. A similar contradiction arises if a branch above P belongs to F_2 .

If $S^0(W)$ contains no trees there is nothing to prove. Otherwise let

$$F \equiv \alpha_\lambda^* F_1(\alpha, \alpha') \equiv \alpha_\lambda^* W_1(a, a'),$$

where

$$\alpha_\lambda^* \equiv AB, \quad W_1 \equiv B'U(a, a'),$$

and the last letter in A is not the inverse of the first letter in U . From the preceding paragraph it follows that the last vertex of α_λ^* either precedes the base of T_1 or is a leaf of T_1 . In the first case the theorem follows immediately from the hypothesis of the induction. In the second case let α_μ^* be the second word in F . From the hypothesis of the induction it follows that every tree in $S^0(W_1)$ is a stump, and if T_1 were not a stump α_μ^* would be completely absorbed by B and the beginning of the third word in F , contradicting the assumption that

⁹ By a segment we shall always mean a non-singular segment.

(α) is reduced (N). Therefore T_1 is a stump, α_μ^* is of the form $B'C$, where $l(B) \leq l(C)$, and T_1 spells out B , read from base to leaf. Since (α) is reduced (N) we have

$$l_0 = l(A) \geq l(B) = h_1,$$

$$l_1 + h_2 \geq l(C) \geq h_1,$$

and the remaining inequalities (2.1a) follow from the hypothesis of the induction, applied to $S^0(W_1)$.

If $l_\rho = 0$ the word starting at the leaf of T_ρ would finish at the leaf of $T_{\rho+1}$ and would be absorbed into its two neighbours, contradicting the assumption that (α) is reduced (N). Therefore $l_\rho > 0$, and the theorem is established.

Two important results¹⁰ are obvious consequences of this theorem, namely:

If the set (α) is reduced (N) it is independent,¹¹ and:

If (α) is reduced (N) and $\Gamma = G$ each α is an α^ .*

It also follows¹² that $l(W^0) \geq l(\xi)$, where ξ is any α^* in

$$F(\alpha, \alpha') = W^0(a, a').$$

For let the segment representing ξ occur after the leaf of T_ρ and before the leaf of $T_{\rho+1}$. Then $l(\xi) \leq h_\rho + l_\rho + h_{\rho+1}$, and from (2.1a) we have

$$l_0 + \dots + l_{\rho-1} \geq h_\rho, \quad h_{\rho+1} \leq l_{\rho+1} + \dots + l_q.$$

Therefore

$$l(W^0) = l_0 + \dots + l_q \geq h_\rho + l_\rho + h_{\rho+1} \geq l(\xi).$$

If $l(W^0) = l(\xi)$, ξ being one of the longest words in F , it follows that $l(\xi) = h_\rho + l_\rho + h_{\rho+1}$ and that $h_\sigma + l_\sigma = h_{\sigma+1}$ if $\sigma < \rho$, $h_\sigma = l_\sigma + h_{\sigma+1}$ if $\sigma > \rho$. Let this be the case and let $\xi_1, \dots, \xi_s$ be the longest words in F , all of equal length, and let $\eta_1, \dots, \eta_t$ be the remaining words, if there are any, each being shorter than the ξ 's. I say that

$$F = F_1(\eta, \eta')F_0(\xi, \xi')F_2(\eta, \eta'),$$

where $F_0 = \xi_\rho^*$ or $\xi_\rho' \xi_\sigma$. For if F begins and ends with a ξ^* the assertion follows from (2.1b) and the fact that

$$l_0 \geq \frac{1}{2} l(\xi), \quad l_q \geq \frac{1}{2} l(\xi)$$

if $q > 0$, while $\epsilon_1 = -1$ and $\epsilon_2 = 1$ if

$$F \equiv \xi_\rho^{*1} \xi_\sigma^{*1},$$

¹⁰ J. N., theorem 1, and *Math. Annalen*, 79 (1919), 269–72.

¹¹ This means that $W(a, a') \neq 1$, where $W(a, a') \equiv F(\alpha, \alpha')$, if $F \neq 1$ and F is reduced (E, α).

¹² J. N., p. 91.

since the last half of each ξ is an isolated ending. If F begins or ends with an η^e we have $l_0 = h_1$ or $l_q = h_q$ and, removing the extreme η^e , the assertion follows from induction on the number of words in F . It also follows that each η^e in F_1 is an η' and each η^e in F_2 is an η , since the last half of any η with an even number of letters is an isolated ending.

By a *level transformation* we shall mean either a transformation of the form $(\alpha) \rightarrow (\alpha^*)$, where (α^*) is any independent set of elements among $\alpha_1, \dots, \alpha_m, \alpha'_1, \dots, \alpha'_m$, or a transformation of the form (1.1) such that $l(\alpha_\lambda^*) = l(\alpha_\lambda)$. Thus $\alpha_\lambda^* = \alpha_\lambda \alpha_\rho$ is a level transformation if, and only if, $\alpha_\lambda \equiv AB$, $\alpha_\rho \equiv B'C$, where $l(B) = l(C)$ and AC is reduced (E, a) . Notice that the inverse of a level transformation is a level transformation. We shall describe (α) as a *minimal set* if

$$L(\alpha) = \sum_{\lambda=1}^m l(\alpha_\lambda)$$

is a minimum among all sets of generators for Γ . From the rule given by Nielsen for reduction (N) it is clear that the reduction of any minimal set to one which is reduced (N) consists of a sequence of level transformations, and from J. N., theorem 6, it follows that any reduced system (N) is minimal.

THEOREM 2. *Any two minimal sets of generators for Γ are interchangeable by level transformations.*

Let (α) and (β) be any two minimal sets of generators for Γ . From the preceding paragraph it follows that no generality is lost in assuming (α) and (β) to be reduced (N) . Then they both contain the same number of elements, say m , and, assuming them to be ordered so that

$$l(\alpha_\lambda) \leq l(\alpha_\mu), \quad l(\beta_\lambda) \leq l(\beta_\mu)$$

if $\lambda < \mu$, we have ¹³

$$l(\alpha_\lambda) = l(\beta_\lambda) = k_\lambda \quad (\lambda = 1, \dots, m).$$

If $m = 0$ the theorem is vacuous, and we shall prove it by induction on m . Let k_s be the first of $k_1, \dots, k_m$ which is the same as k_m . If $s > 1$ and $\lambda < s \leq \rho$ we have $l(\beta_\lambda) < l(\alpha_\rho)$, $l(\alpha_\lambda) < l(\beta_\rho)$ and it follows from a previous result that $(\beta_1, \dots, \beta_{s-1})$ depend only on $(\alpha_1, \dots, \alpha_{s-1})$ and vice versa. Any sub-set of a minimal set is obviously minimal and if $s > 1$ it follows from the hypothesis of the induction that $(\beta_1, \dots, \beta_{s-1}) \rightarrow (\alpha_1, \dots, \alpha_{s-1})$ by a sequence of level transformations. Moreover, it follows from a previous argument that

$$(\alpha_1, \dots, \alpha_{s-1}, \beta_s, \dots, \beta_m) \rightarrow (\alpha_1, \dots, \alpha_{s-1}, \gamma_s, \dots, \gamma_m)$$

by level transformations, where

$$\gamma_\lambda = \alpha_{\rho_\lambda}^e \text{ or } \alpha_{\rho_\lambda}' \alpha_{\sigma_\lambda} \quad (s \leq \lambda, \rho_\lambda, \sigma_\lambda \leq m).$$

¹³ J. N., p. 91.

If $s = 1$ it follows from the same argument that $(\beta) = (\gamma)$ is related to (α) in this way.

If each γ is an α^e a further level transformation $(\gamma) \rightarrow (\alpha_s, \dots, \alpha_m)$ carries $(\alpha_1, \dots, \alpha_{s-1}, \gamma_s, \dots, \gamma_m)$ into (α) . Otherwise we can arrange the notation so that

$$\begin{aligned}\gamma_\lambda &= \alpha_\lambda & (\lambda = s, \dots, t) \\ \gamma_\mu &= \alpha'_{\rho_\mu} \alpha_{\sigma_\mu} & (\mu = t+1, \dots, m),\end{aligned}$$

with the convention $t = s - 1$ if no γ is an α^e . If $\rho_\mu > t, \sigma_\mu > t$ for every value of μ the determinant of the linear transformation

$$\begin{aligned}\gamma_\lambda &= \alpha_\lambda \\ \gamma_\mu &= \alpha_{\sigma_\mu} - \alpha_{\rho_\mu} = -\alpha_{\rho_\mu} + \alpha_{\sigma_\mu},\end{aligned}$$

calculated to the modulus 2, would be zero. Therefore some γ is an α , and $\rho_\mu \leq t$ or $\sigma_\mu \leq t$ for at least one value of μ , say $\gamma_{t+1} = \alpha'_t \alpha_\sigma$. Then $\sigma > t$, or the set (γ) would not be independent, and we may suppose $\sigma = t + 1$. Applying the level transformation

$$\gamma_{t+1}^* = \gamma_t \gamma_{t+1} = \alpha_{t+1}$$

to (γ) , the theorem follows from a second induction on $m - t$.

It is a straightforward matter to write down all the sets obtainable from a given minimal set (α^0) , by sequences of level transformations. For let $(\alpha^1), \dots, (\alpha^{k_1})$ be all the sets obtained from (α^0) by single level transformations. Omitting those which already appear in the sequence, let $(\alpha^{k_1+1}), \dots, (\alpha^{k_2})$ be the totality of sets obtained from each of $(\alpha^1), \dots, (\alpha^{k_1})$ by single level transformations, and let $(\alpha^{k_2+1}), \dots, (\alpha^{k_{s+1}})$ be similarly obtained from $(\alpha^{k_{s-1}+1}), \dots, (\alpha^{k_s})$ ($s = 1, 2, \dots$). This process will come to an end after a number of steps which is bounded above by $(2p)^{L(\alpha^0)}$, since this number exceeds the number of sets (ξ) , for which $L(\xi) = L(\alpha^0)$. If (ξ) is any set derived from (α^0) by a sequence of level transformations let $(\xi^{\rho+1})$ be derived from (ξ^ρ) by a level transformation, where $\rho = 0, \dots, N$, $(\xi^0) = (\alpha^0)$, $(\xi^N) = (\xi)$, and¹⁴ $(\xi^\rho) \neq (\xi^\sigma)$ if $\rho \neq \sigma$. Then it follows from induction on ρ that (ξ^ρ) is one of $(\alpha^{k_{\rho-1}}), \dots, (\alpha^{k_\rho})$, where $k_0 = 1$. Therefore the sequence $(\alpha^0), (\alpha^1), \dots$ contains every set which can be obtained from (α^0) by a sequence of level transformations. Thus a reduced set (β) will be contained in this sequence if it is equivalent to (α^0) .

'In general' no level transformation of the form (1.1) is applicable to a reduced set (α) . In such a case we have only to compare (β, β') with the various permutations of (α, α') to discover whether or not (α) is equivalent to a given reduced set (β) .

Following the analogy with linear dependence, one might enquire whether or

¹⁴ If $(\xi^\rho) = (\xi^\sigma)$ where $\sigma > \rho$ we should simply omit $(\xi^{\rho+1}), \dots, (\xi^\sigma)$ from the sequence.

no it is always possible to select an independent basis for Γ from a basis between whose elements there are one or more identical relations. The example

$$\alpha = a^2ba'b', \quad \beta = b^2ab'a', \quad \gamma = aba'b'$$

shows that the answer is in the negative. For $a = \alpha\gamma'$, $b = \beta\gamma$, but no two of (α, β, γ) generate the free group $G(a, b)$.

3. We now consider the equivalence of ordered sets of elements under the automorphisms of G . A simple transformation, operating on a set of words (W) , true or cyclic, will mean the same as in §1 of W. G., referred to in §1 above, except that we shall include $a_i \rightarrow a_i'$ among the simple transformations. By a *transformation of type T* , or a *T -transformation*, we shall mean a transformation of the form¹⁵

$$(3.1) \quad a_i^{\epsilon_i \rho} \rightarrow a_i^{\epsilon_i \rho} a_k^{\epsilon_i} \quad (k \text{ fixed, } \rho = 1, \dots, r, i_\rho \neq k),$$

followed by what we shall call a *complete elementary reduction*, meaning a transformation by elementary reductions to words which are reduced (E). Here it is to be understood that $a_i^{\epsilon_i \rho} \neq a_i^{\epsilon_i \sigma}$ if $\rho \neq \sigma$, but we allow $i_\rho = i_\sigma$, $\epsilon_\rho = -\epsilon_\sigma$, in which case

$$a_{i_\rho} \rightarrow a_k^{-\epsilon_i} a_{i_\rho} a_k^{\epsilon_i}.$$

If (W) is reduced (E) and if (W^*) is obtained from (W) by a T -transformation we shall describe the latter as a *reduction (T)*, or as a *T -reduction*, if

$$L(W^*) < L(W),$$

and as a *level transformation* if

$$L(W^*) = L(W).$$

We shall also describe an automorphism of the form¹⁶

$$a_i \rightarrow a_j^{\epsilon_i \epsilon_j}$$

as a level transformation. A set to which no T -reduction is applicable will be described as *minimal*, or as minimal (T) if confusion with the minimal sets of §2 is to be feared. The latter may then be described as minimal (N). The total number of T -transformations is finite and one can find out if a given one, applied to a given set, is a reduction or a level transformation or neither. Also the sets (W^*) , such that $(W) \rightarrow (W^*)$ by level transformations, can be written down by a process similar to the one described above.

We shall also introduce a new use of the term elementary transformation. By an *elementary transformation* we shall now mean any one of the following:

¹⁵ Cf. W.G., §2.

¹⁶ Such a transformation is a product of simple transformations (Nielsen, *Math. Annalen*, 79 (1919), 269–72).

1. The identical transformation,
2. An elementary reduction or insertion, these being defined as usual,
3. A transformation of the form¹⁷

$$W_\lambda \rightarrow x' W_\lambda x \quad (\lambda = 1, \dots, m),$$

where W_λ may be true words provided x is the same for each value of λ ,

4. The resultant of any sequence of these.

It appears that the equivalence test is analogous to the one described at the end of §2. If we reduce (W) to a minimal set (W^0) by a random sequence of T -reductions, all the minimal sets which are equivalent to (W) can be obtained from (W^0) by level transformations.

To prove this we shall use the model for (W) described in W. G. §3, but will denote the 2-spheres in M by Σ_i instead of Γ_i . Let $\Sigma_1^0, \dots, \Sigma_p^0$ be any set of oriented, non-intersecting surfaces in M , and let $w_1, \dots, w_m$ be any set of oriented circuits¹⁸ in M which, with the 2-spheres Σ , represent the words (W) . Then the surfaces Σ^0 , together with the circuits w , represent a set of words (W^0) , a positive intersection $\Sigma_i \cdot w_\lambda$ being marked a_i , and a negative intersection $\Sigma_i \cdot w_\lambda$ being marked a'_i . Thus the substitution $\Sigma_i \rightarrow \Sigma_i^0$, which consists of replacing Σ_i by Σ_i^0 in the machine for spelling out words, determines a transformation $W_\lambda \rightarrow W_\lambda^0$.

Now let the surfaces $\Sigma_1^0, \dots, \Sigma_p^0$ be non-singular 2-spheres, no linear combination of which bounds, and let $S_1^0, \dots, S_q^0$ be the connected surfaces in the model R , obtained by cutting M along $\Sigma_1, \dots, \Sigma_p$ (cf. W. G. §3), which correspond to regions on the 2-spheres Σ^0 , the surfaces S^0 being bounded, in general, by circuits on the 2-spheres C_i and C'_i . Using a method due to J. Singer,¹⁹ as adapted in W. G. (p. 54), we shall prove two lemmas, of which the first is:

LEMMA 1. *The 2-spheres Σ^0 are similar to the 2-spheres Σ in that M is transformed into a punctured sphere²⁰ R^0 , by cutting along $\Sigma_1^0, \dots, \Sigma_p^0$.*

To prove this we modify Singer's argument as follows. We replace his diagram Δ' by our model R , his circuits e by our 2-spheres Σ^0 and his circuits e' by our 2-spheres Σ . We replace the points P_1 and P_2 of the third paragraph on his p. 106 by a circuit belonging to some intersection $\Sigma_i^0 \cdot \Sigma$, which bounds a 2-element on Σ_i^0 containing no other circuit belonging to an intersection $\Sigma_i^0 \cdot \Sigma$. We also define moves analogous to his IA, IB and IC. A move of type IA changes the orientation of a 2-sphere Σ and determines a simple transformation of the form $a_i \rightarrow a'_i$. Let C^* be a 2-sphere in R which either

¹⁷ On p. 50 of W. G. it is shown that such a transformation is the product of simple transformations.

¹⁸ As in W. G., all the chains referred to will have general intersections, relative to slight deformations, unless the contrary is stated.

¹⁹ *Trans. American Math. Soc.*, 35 (1933), pp. 105-6.

²⁰ By a punctured sphere is meant a bounded manifold which is equivalent to a 3-sphere from which a (finite) number of 3-elements have been removed, no two of the latter having a common vertex.

1. separates some $C_i^e (= C_i \text{ or } C'_i)$ and no other 2-sphere C_i^e from C_i^{-e} , or
2. separates $C_i^{e,j}$ and precisely one other 2-sphere $C_i^{e,i}$, from $C_i^{-e,i}$.

We cut R along C^* and re-attach the two regions so formed by uniting corresponding vertices of C_i and C'_i . This is a move of type IB in the first case and of type IC in the second.

Let U be the region, bounded by C^* and C_i^e , which is detached from R in a move of type IB, and let $C^{*'} be the side of C^* incident with U . Let U^* be the region into which U is transformed²¹ by uniting C_i with C'_i , let $R_1 = R - U$, $R^* = R_1 + U^*$ and let E be a 3-element bounded by $C^{*'}$ but having no internal component in common with R . Then R_1 and U are punctured spheres, $U^* + E$ is a 3-element, $R_1 + (U^* + E)$ is a punctured sphere and $R^* = (R_1 + U^* + E) - E$ is a punctured sphere. Thus a move of type IB transforms R into another punctured sphere. A similar argument shows that a move of type IC also transforms R into another punctured sphere, and Singer's argument shows that one can, by a series of these moves, transform R into a punctured sphere R^0 , whose boundary 2-spheres correspond in pairs to the 2-spheres Σ^0 . Thus lemma 1 is established.$

The second lemma is:

LEMMA 2. *The set of words (W^0) is equivalent to the set (W) .*

First let $R^0 = R$, that is to say let each Σ^0 be a Σ . Then we can transform (W) into (W^0) by a change of notation

$$a_j \rightarrow a_i^e j.$$

Now let R be transformed into R^0 by a single move of type IB or type IC, and first let this be of type IB. Rearranging the notation, if necessary, let $\Sigma_i^0 = \Sigma_i$ if $i \neq j$. Then S_j^0 , the image of Σ_j^0 in R , is the 2-sphere C^* , and we may assume that S_j^0 separates C_j , rather than C'_j , from the rest of the 2-spheres C_i^e , and that Σ_j^0 is oriented so that the positive side of S_j^0 is the one remote from C_j . Then the transformation $\Sigma_j \rightarrow \Sigma_j^0$, with due regard to orientation, can be regarded as the result of an isotopic deformation of Σ_j into Σ_j^0 . Such a deformation is the resultant of deformations of the form

$$\Sigma \rightarrow \Sigma + \hat{E} \quad (\text{mod. } 2),$$

where $\hat{E}$ is the boundary of a 3-element E , which meets Σ in a 2-element f , on $\hat{E}$, and nowhere else. After a suitable subdivision and slight deformation of the circuits w , the intersections $\Sigma.w$ will be general after each stage of the deformation $\Sigma_j \rightarrow \Sigma_j^0$. Thus if (W) consists of cyclic words, or if (W) consists of true words and none of the 3-elements E contains O , the deformation $\Sigma_j \rightarrow \Sigma_j^0$ will determine an elementary transformation of (W) , since the intersections $\Sigma.w$ appear and disappear in consecutive pairs, those in each pair differing in sign. If (W) consists of true words and if some E contains O we modify the

²¹ The transformation from U to U^* consists of replacing the vertices of C_i^e by the corresponding vertices of C_i^{-e} .

deformation as follows. Let E^0 be a 3-element in E whose boundary meets $\hat{E}$ in a 2-element contained in f and nowhere else, and such that $\hat{E}^0$ meets each segment of w issuing from O in a single vertex, and otherwise does not meet w . For example, after a suitable subdivision of E , we may take

$$E^0 = N(s, E),$$

where s is a segment joining O to f and $N(L, K)$ stands for the aggregate of components in a complex K which meet L . Then we take the deformation

$$\Sigma \rightarrow \Sigma + \hat{E}$$

in two stages, namely

$$\Sigma \rightarrow \Sigma + \hat{E}^0$$

and

$$\Sigma + \hat{E}^0 \rightarrow \Sigma + \hat{E}^0 + (\hat{E} + \hat{E}^0).$$

The first stage determines an elementary transformation of the form

$$W_\lambda \rightarrow a_i^\epsilon W_\lambda a_j^{-\epsilon}$$

and the second stage also determines an elementary transformation. Thus in all cases

$$\Sigma \rightarrow \Sigma + \hat{E},$$

and hence the complete deformation

$$\Sigma_j \rightarrow \Sigma_j^0,$$

determines an elementary transformation of the words (W).

Now let R be transformed into R^0 by a single move of type IC. As before let S_j^0 cover C_j positively and let the detached region be bounded by S_j^0 , C_i and C_j . Let μ stand for the move $R \rightarrow R^0$ and let μ^* be a move of type IC in which R is cut along a 2-sphere C^* , separating C_i and C_j from S_j^0 . Then μ is the resultant of μ^* , followed by a move of type IB. After a suitable subdivision of R let

$$C^* = N(C_i + s + C_j, R),$$

where s is a segment joining C_i to C_j in the region bounded by C_i , C_j and S_j^0 , and the positive side of C^* is the one remote from C_j . According to an argument in W. G. (p. 54) the move μ^* determines the simple transformation $a_i \rightarrow a_i a_j$ of the words (W), C^* being eventually renamed C_j . Therefore μ determines the resultant of the simple transformation $a_i \rightarrow a_i a_j$ followed by an elementary transformation.

In the general case the lemma now follows from induction on the number of moves in the transformation $R \rightarrow R^0$.

Let n_i be the number of times a_i and a'_i occur in (W) . Then the circuits w pierce Σ_i in n_i vertices, and we shall prove the lemma:

LEMMA 3. *If (W) is a minimal set (T) and if C^* is a 2-sphere in R which separates C_k from C'_k then*

$$n^* \geq n_k,$$

where the circuits w pierce Σ^* , the image of C^* , in n^* vertices.

Let C^* separate C_k^e , and also $C_{i_1}^{e_1}, \dots, C_{i_r}^{e_r}$ say, from C_k^{-e} , with the convention that $r = 0$ if C^* is isotopic to C_k^e . If $r = 0$ the transformation of (W) determined by $\Sigma \rightarrow \Sigma^*$ is an elementary transformation. Otherwise let s_p be a segment in the region bounded by $C_k^e, C_{i_1}^{e_1}, \dots, C_{i_r}^{e_r}$ and C^* , joining C_k^e to $C_{i_p}^{e_p}$, and, after a suitable sub-division, let

$$C^{**} = N(C_k^e + C_{i_1}^{e_1} + \dots + C_{i_r}^{e_r} + s_1 + \dots + s_r, R),$$

oriented so that the positive side of C^{**} is the one remote from C_k . Let Σ^{**} be the image of C^{**} in M . As in the case of a single move of type IC (i.e. the case $r = 1$), it follows, from an obvious extension of the argument in W. G., that the transformation of (W) determined by $\Sigma \rightarrow \Sigma^{**}$ is the transformation (3.1). Therefore the transformation determined by $\Sigma \rightarrow \Sigma^*$ is (3.1) followed by an elementary transformation.

In either case, let

$$\Sigma_i^0 = \Sigma_i \text{ if } i \neq k, \quad \Sigma_k^0 = \Sigma^*,$$

and let (W^0) be the set of words determined by the circuits w and the 2-spheres Σ^0 . Then

$$L(W^0) = L(W) + n^* - n_k,$$

and we have only to show that $L(W) \leq L(W^0)$.

If $r = 0$, $L(W) \leq L(W^0)$ since $(W) \rightarrow (W^0)$ is an elementary transformation and (W) is reduced (E) . If $r > 0$ let (W^*) be the set of words derived from (W) by the T -transformation consisting of (3.1) followed by a complete elementary reduction. Then $L(W^*) \leq L(W^0)$, since (W^0) is derived from (W) by (3.1) followed by an elementary transformation which may or may not be a complete elementary reduction. Since (W) is minimal $L(W) \leq L(W^*)$. Therefore in either case

$$L(W) \leq L(W^0)$$

and the lemma is established.

Let (W) and (W^0) be two sets of words represented by the intersections of a set of circuits w^0 with sets of independent 2-spheres Σ and Σ^0 . In W. G. (p. 53) I described a method for replacing Σ^0 by a set of surfaces having the same intersections with w^0 and having normal intersections with the 2-spheres Σ . That is to say each of the new surfaces S^0 meets a given one of the 2-spheres C_i^e in a single circuit, if at all. Here we shall need the result stated in the lemma:

LEMMA 4. *The circuits w^0 can be replaced by a set of circuits whose intersections*

with Σ also represent the words (W) and, at the same time, Σ^0 can be replaced by a set of independent 2-spheres having normal intersections with Σ , whose intersections with the new circuits represent the words (W^0).

Unless the intersections $\Sigma \cdot \Sigma^0$ are already normal there is a surface S_β^0 such that two circuits of $\hat{S}_\beta^0$ are connected by a segment pq , on some C_i^* , which does not meet any other circuit belonging to an²² $\hat{S}^0$. Let E_1 and E_2 ($E_2 \supset q$) be the 2-elements on C_i^* which are bounded by these circuits of $\hat{S}_\beta^0$ and which do not contain pq . After a suitable sub-division let E_2^* be the 2-element on the boundary of $N(E_2, R)$ which is inside R and is bounded by the circuit on the boundary of $N(\hat{E}_2, S_\beta^0)$ which is inside R , and let pq^* be a segment on S_β^0 joining p to a vertex q^* on E_2^* . Let T be the solid tube in R bounded by

$$E_2 + E_2^* + N(\hat{E}_2, S_\beta^0).$$

We shall replace T by a 3-element T^* , defined as follows. Let R be sub-divided again in such a way that

$$F = N(E_2, T) \text{ and } F^* = N(E_2^*, T)$$

are 3-elements which do not meet each other. Let t consist of all the solids in R which meet $qp + pq^*$ and lie on the same side²³ of S_β^0 as pq . Then we assume the second sub-division of R to be such that

$$T^* = F + t + F^*$$

is a 3-element. We map T semi-linearly on T^* in such a way that each vertex of E_2 and E_2^* corresponds to itself. If one of the segments in R corresponding to a segment of some w pierces E_2 it will also pierce E_2^* without leaving T , because of our understanding that all intersections are general. Similarly any portion of an S^0 which lies inside T is a cylinder bounded by circuits in E_2 and E_2^* . We replace these segments and cylinders²⁴ by their images in T^* , and S_β^0 by

$$S_\beta^0 + \hat{T} + \hat{T}^*,$$

which is a surface meeting C_i^* in the same circuits as S_β^0 and also in a strip bordering pq , since addition is mod. 2. The corresponding alteration in M substitutes other 2-spheres for any of the 2-spheres Σ^0 which have been modified. For we have not altered the topology of these surfaces by substituting cylinders internal to T^* for cylinders internal to T . In altering T itself we have merely removed the cylinder bounded by $\hat{E}_2$ and $\hat{E}_2^*$, deformed $\hat{E}_2^*$ by removing a strip on S_β^0 bordering pq^* and then restored a cylinder bounded by $\hat{E}_2$ and the image of $\hat{E}_2^*$ in the deformation. Since all the chains have general intersections neither the

²² W. G., p. 53.

²³ The surfaces S^0 are two-sided, since they correspond to regions on the 2-spheres Σ^0 , and E_2 and E_2^* lie on the same side of S_β^0 , pq and t lying on the other side.

²⁴ Though addition is mod. 2 we shall sometimes use the minus sign to make the formulae more explicit.

first cylinder nor the strip bordering pq^* meets any of the segments corresponding to the circuits w . Any segments which previously ran through T are now replaced by segments in T^* with the same end points. Thus the circuits w and the 2-spheres Σ^0 are replaced by new sets of circuits and 2-spheres, say w^* and Σ^* , such that the intersections $\Sigma \cdot w^*$ and $\Sigma^* \cdot w^*$ are the same as $\Sigma \cdot w$ and $\Sigma^0 \cdot w$ respectively.

Any one of the cylinders on a surface S^0 which ran through T , together with 2-elements on E_2 and E_2^* , bounded a 3-element T_λ , in T , and the cylinder in T^* which replaces it, together with the same 2-elements, bounds a 3-element T_λ^* , in T^* . Let $T_0 = T$, $T_0^* = T^*$ and let τ_λ and τ_λ^* be the respective images of T_λ and T_λ^* in M ($\lambda = 0, \dots, n$). Then either

$$\Sigma_k^* = \Sigma_k^0$$

or

$$\Sigma_k^* = \Sigma_k^0 + \hat{\tau}_0 + \hat{\tau}_0^* + \dots + \hat{\tau}_n + \hat{\tau}_n^*.$$

Therefore

$$\Sigma_k^* \sim \Sigma_k \quad (k = 1, \dots, p),$$

and the 2-spheres Σ^* are independent.

The circuits $\hat{E}_1$ and $\hat{E}_2$ may now be united by a slight deformation which carries the strip bordering pq through C_i^+ into the neighbourhood of C_i^- , and the lemma follows from induction on the total number of circuits in the boundaries²⁵ $\hat{S}_1^0, \dots, \hat{S}_q^0$.

As a final lemma we prove the converse of lemma 2, restricting ourselves to the case where (W) and (W^0) are reduced (E) .

LEMMA 5. *Any two equivalent sets of words (W^0) and (W) , both reduced (E) , can be simultaneously represented by the intersections of a single set of circuits w , with sets of independent 2-spheres Σ^0 and Σ .*

Let (W^0) and (W) be equivalent sets of words, both reduced (E) , and let (W^0) be represented in M by the intersections of a set of circuits w^* , with a set of independent 2-spheres Σ^0 . From an argument in W. G., §1, it follows that the transformation $(W^0) \rightarrow (W)$ can be exhibited in the form

$$(W) = Ut_\mu \dots t_1(W^0),$$

where t_ρ is a simple transformation of the form $x \rightarrow xy$, not followed by any elementary reductions, and U consists only of elementary reductions, (W) being reduced (E) , and transformations of the type $x \rightarrow x'$. As in W. G. (p. 54), it follows from induction on μ that

$$(W^*) = t_\mu \dots t_1(W^0)$$

²⁵ From the existence of the original segment pq , it follows that the relation 'having normal intersection with' is symmetric between Σ and Σ^0 .

is represented in M by the intersections of the original circuits w^* with a set of independent 2-spheres Σ . In the presence of lemma 4 we lose no generality in assuming the intersections $\Sigma \cdot \Sigma^0$ to be normal, which we accordingly do.

As in W. G. (p. 54), where (W^0) is an independent set of letters, we have to show that the reduction $(W^*) \rightarrow (W)$ can be copied by a deformation of the circuits w^* into a set whose intersections with Σ^0 and Σ represent (W^0) and (W) respectively. Let PQ be the segment in R corresponding to a syllable xx' in (W^*) , say $a_i a'_i$. Since the intersections $\Sigma^0 \cdot \Sigma$ are normal, P and Q are joined by a segment s , on C_i , which does not meet any one $\tilde{S}_\beta^0$ more than once, and PQ can be deformed²⁶ into s in R . As we copy this deformation in M the intersections $w^* \cdot \Sigma^0$ may be modified. If so it follows from an argument similar to one used in proving lemma 2 that the new intersections $w^* \cdot \Sigma^0$ will represent a set of words which are derived from (W^0) by elementary reductions and insertions. But an argument used in W. G. shows that the number of intersections $s \cdot \tilde{S}^0$ does not exceed the number of intersections $PQ \cdot S^0$. That is to say, the number of intersections $w^* \cdot \Sigma^0$, and therefore the number of letters in (W^0) , is not increased. Since (W^0) was originally reduced (E) it follows that the set of circuits w^* continues to represent (W^0) after the deformation. As in W. G., the elementary reduction which consists of omitting $a_i a'_i$ may now be copied in R by deforming s through C_i into the neighbourhood of C'_i . Repeating this process, w^* may be replaced by a set of circuits w , whose intersections with Σ^0 and Σ represent (W^0) and (W) respectively. Thus the lemma is established.

We are now ready to prove:

THEOREM 3. *Any two equivalent minimal sets (T) are interchangeable by level T -transformations.*

The inverse of a T -transformation, applied to a set which is reduced (E) , is a T -transformation. Therefore the inverse of a level transformation is a level transformation and it is enough to show that two equivalent minimal sets (W^0) and (W) can be identified by level transformations applied indiscriminately to either or both of them.

Let (W^0) and (W) be represented in M by the intersections of w with independent sets of 2-spheres Σ^0 and Σ . First let us assume that each Σ^0 either coincides with some Σ or does not meet any Σ . Let the notation be such that

$$\Sigma_1^0, \dots, \Sigma_k^0$$

do not meet any Σ , while

$$\Sigma_\rho^0 = \Sigma_\rho \text{ if } \rho > k$$

(possibly $k = p$, in which case no Σ^0 meets a Σ). If k is zero $(W) = (W^0)$, and we shall use induction on k .

We have

$$\Sigma_\rho^0 \sim r_{\rho j} \Sigma_j \quad (\text{summed for } j = 1, \dots, p),$$

²⁶ We need not concern ourselves with the fact that PQ may cross itself or other segments of $g(w^*)$ during this deformation.

where $r_{\rho i}$ are calculated mod. 2. Also $|r_{\rho\sigma}| = 1$ ($\rho, \sigma = 1, \dots, k$), since the Σ^0 's are independent. Rearranging the notation, if necessary, we may suppose $r_{11} = \dots = r_{kk} = 1$. Then S_ρ^0 , the image of Σ_ρ^0 in R , separates C_ρ from C'_ρ , and since (W) is a minimal set it follows from lemma 3 that

$$n_\rho^0 \geq n_\rho,$$

where n_i^0 is the number of intersections $\Sigma_i^0 \cdot w$. Since $n_i^0 = n_i$ if $i > k$ we have

$$L(W^0) = n_1^0 + \dots + n_p^0 \geq n_1 + \dots + n_p = L(W).$$

Similarly

$$L(W) \geq L(W^0)$$

and it follows that

$$L(W) = L(W^0).$$

Therefore²⁷ $n_\rho^0 = n_\rho$. We have seen that the transformation of the words (W) determined by the substitution $\Sigma_k \rightarrow \Sigma_k^0$ can be exhibited either as an elementary transformation U , or as a transformation of the form (3.1) followed by an elementary transformation U . In the first case U is the identity, since $n_k^0 = n_k$ and the words (W) are reduced (E) . In the second case, U is a complete elementary reduction since the set (W) is minimal. Otherwise the T -transformation determined by (3.1) would be a T -reduction, since $n_k^0 = n_k$ and any elementary reduction applied after U would reduce the number of letters in the new words (W) . Therefore $\Sigma_k \rightarrow \Sigma_k^0$ is a T -transformation. Therefore it is a level transformation and this special case of the theorem follows from induction on k .

Returning to our general rule that all intersections are to be general relative to slight deformations, we shall complete the proof by induction on the number of circuits in which Σ and Σ^0 intersect. The process of normalization used in proving lemma 4 reduces the number of circuits in the intersections $\Sigma \cdot \Sigma^0$. Therefore we may assume the latter to be normal.

Let S_ρ^0 be any bounded S^0 and let

$$\hat{S}_\rho^0 = c + c'_1 + \dots + c'_r$$

where c is on C_i^ϵ and c'_ρ is on $C_i^{-\epsilon, \rho}$ (possibly $C_i^{-\epsilon, \rho} = C_i^{-\epsilon}$ for some ρ). Let E be one of the 2-elements into which C_i^ϵ is separated by c , let E'_ρ be one of the 2-elements into which $C_i^{-\epsilon, \rho}$ is separated by c'_ρ and let

$$F = S_\rho^0 + E'_1 + \dots + E'_r.$$

Let $M(K)$ stand for the image in M of any complex K in R , let Σ_j^0 be the Σ^0 containing $M(S_j^0)$ and let E^0 be the 2-element on Σ_j^0 which is bounded by $M(c)$ and contains $M(S_j^0)$. Using induction on the number of circuits belonging to

²⁷ Cf. J. N., p. 91.

$\Sigma \cdot \Sigma_j^0$ which are inside E^0 , we shall define the *proper* one of the two 2-elements into which C_i^* is separated by c . If

$$(3.2) \quad M(C_i^* - E + F) \sim x_i \Sigma_i, \quad M(E + F) \sim y_i \Sigma_i$$

(summed over i), we have $x_i + y_i = 1$, the i being the same as in C_i^* . If E^0 does not meet any Σ except in its boundary, $\hat{S}_\beta^0$ consists of a single circuit. Otherwise we choose E'_ρ on $C_i^{-1}{}_\rho^*$ such that its image E_ρ , on $C_i^*_\rho$, is the proper 2-element bounded by c_ρ , the image of c'_ρ . In either case the proper 2-element E , on C_i^* , is to be the one such that the y_i of (3.2) is zero and $x_i = 1$.

If k is the number of intersections $M(E) \cdot w$, E being the proper 2-element of the last paragraph, and if k^0 is the number of intersections $E^0 \cdot w$, I say that

$$k \leq k^0.$$

For let k_ρ be the number of intersections $M(E_\rho) \cdot w$ and k_ρ^0 the number of intersections $E_\rho^0 \cdot w$, where E_ρ^0 is the 2-element in E^0 bounded by $M(c_\rho)$. Then

$$(3.3) \quad k^0 = s^0 + k_1^0 + \dots + k_r^0,$$

where s^0 is the number of intersections $M(S_\beta^0) \cdot w$. Since

$$M(C_i^* - E + F) \sim \Sigma_i + \dots$$

it follows that the 2-sphere $C_i^* - E + F$, after a slight deformation in R , separates C_i from C_i' . The number of intersections $M(C_i^* - E + F) \cdot w$ is $n_i - k + s^0 + k_1 + \dots + k_r$, and it follows from lemma 3 that

$$(3.4) \quad k \leq s^0 + k_1 + \dots + k_r.$$

If $\hat{S}_\beta^0$ consists of the single circuit c , $k_\rho = 0$ and $k^0 = s^0$. Otherwise, assuming $k_\rho \leq k_\rho^0$, the required result follows from (3.4), (3.3) and induction on the number of circuits of $\Sigma \cdot \Sigma_j^0$ inside E^0 .

Let Σ_i be one of the 2-spheres Σ which meets a 2-sphere in the set Σ^0 . Let c be an 'extreme' circuit belonging to $\Sigma_i \cdot \Sigma^0$, meaning that one of the 2-elements into which Σ_i is separated by c contains no other circuit belonging to $\Sigma_i \cdot \Sigma^0$. Let A be the 2-element in question and let E^0 be the 2-element bounded by c on Σ_j^0 , the Σ^0 which contains c , such that $x_i = 1$ in the homology

$$\Sigma_j^0 - E^0 + A \sim x_i \Sigma_i^0.$$

Now cut M along the 2-spheres Σ^0 to form the model R^0 , referred to in lemma 1. The image in R^0 of $\Sigma_j^0 - E^0 + A$, slightly deformed, is a 2-sphere which separates the two images of Σ_j^0 from each other. The number of intersections $(\Sigma_j^0 - E^0 + A) \cdot w$ is $n_j^0 - k^0 + l$, where k^0 is the number of intersections $E^0 \cdot w$ and l is the number of intersections $A \cdot w$. Therefore

$$l \geq k^0,$$

according to lemma 3. If $l = k^0$ it follows from a previous argument that the substitution $\Sigma_j^0 \rightarrow \Sigma_j^0 - E^0 + A$ determines a level transformation of (W^0) .

Since it reduces the number of intersections $\Sigma \cdot \Sigma^0$ the theorem, in this case, follows from the hypothesis of the induction. If $l > k^0$ let C_i^ϵ be the image in R of the side of Σ_i incident with E^0 . If $R(A)$, the image of A in R , were the proper 2-element bounded on C_i^ϵ by $R(c)$ we should have $l \leq k^0$. Therefore

$$E = C_i^\epsilon - R(A)$$

is the proper 2-element, and if k is the number of intersections $M(E) \cdot w$ we have $k \leq k^0 < l$. That is to say, more than half the intersections $\Sigma_i \cdot w$ lie in A unless the substitution $\Sigma_j^0 \rightarrow \Sigma_j^0 - E^0 + A$ determines a level transformation of (W^0) .

But the 2-element $\Sigma_i - A$ also contains a 2-element A_1 , which is bounded by a circuit of $\Sigma_i \cdot \Sigma^0$ and contains no other circuit of $\Sigma_i \cdot \Sigma^0$. Adding suitable subscripts to indicate that A is replaced by A_1 we see that one or other of the substitutions

$$\Sigma_j^0 \rightarrow \Sigma_j^0 - E^0 + A, \quad \Sigma_{j_1}^0 \rightarrow \Sigma_{j_1}^0 - E_1^0 + A_1$$

determines a level transformation of (W^0) . They both reduce the number of circuits in $\Sigma \cdot \Sigma^0$, and the theorem is established.

From this theorem it is clear that the process described earlier in this section will exhibit the equivalence of equivalent sets. As before, let n_i be the number of times a_i and a'_i appear in (W) and let q be the number of n 's which are not zero. Then $L(W)$, q and n_i are each unaltered by level T -transformations and it follows that these numbers, calculated for a minimal (T) set (W) , are invariants of the family of equivalent sets. They are analogous to $L(\alpha)$, m and $k_\lambda = l(\alpha_\lambda)$, calculated for a minimal (N) set (α) .

4. In this section we indicate how the results of §§2 and 3 can be applied to the study of free factor groups of free groups. Let Γ be the free group generated by $\alpha_1, \dots, \alpha_m$ and let γ be the factor group Γ/Γ_0 , where Γ_0 is the smallest invariant sub-group containing

$$R_1(\alpha, \alpha'), \dots, R_q(\alpha, \alpha').$$

Let $\alpha_\lambda \rightarrow \alpha_\lambda(a, a')$ define a simple isomorphism between γ and the free group G , generated by $a_1, \dots, a_p$. Then

$$(4.1) \quad R_\lambda[\alpha(a, a'), \alpha'(a, a')] = 1,$$

an element $F(\alpha, \alpha')$, of Γ , belongs to Γ_0 if

$$F[\alpha(a, a'), \alpha'(a, a')] = 1,$$

and there are elements $F_i(\alpha, \alpha')$ such that

$$a_i = F_i[\alpha(a, a'), \alpha'(a, a')].$$

Therefore $\alpha_1(a, a'), \dots, \alpha_m(a, a')$ constitute a basis for G , and there is a basis for Γ , say $\beta_1(\alpha, \alpha'), \dots, \beta_m(\alpha, \alpha')$, such that the set consisting of the corresponding elements in G , namely

$$\beta_\lambda(a, a') = \beta_\lambda[\alpha(a, a'), \alpha'(a, a')],$$

is reduced (N). Re-arranging the notation, if necessary, it follows from theorem 1 that

$$(4.2) \quad \begin{cases} \beta_i(a, a') = a_i & (i = 1, \dots, p), \\ \beta_\rho(a, a') = 1 & (\rho = p+1, \dots, m), \end{cases}$$

m exceeding²⁸ p unless the relations $R_\sigma = 1$ are vacuous. Then γ is the subgroup freely generated by $\beta_1, \dots, \beta_p$, and the relations

$$(4.3) \quad S_1(\beta, \beta') = \dots = S_q(\beta, \beta') = 1$$

are equivalent to

$$(4.4) \quad \beta_{p+1} = \dots = \beta_m = 1,$$

where

$$S_\sigma(\beta, \beta') = R_\sigma[\alpha(\beta, \beta'), \alpha'(\beta, \beta')].$$

In particular, let γ be defined by a single relation

$$R(\alpha, \alpha') = 1,$$

where $R \neq 1$ when the word R is reduced (E, α). With the notation used in the last paragraph we see by considering the free Abelian group associated with Γ that $p = m - 1$, since $p < m$ and the relations (4.4) are equivalent to (4.3), with $q = 1$. Moreover it follows from the equivalence of (4.3) and (4.4) that²⁹

$$S(\beta, \beta') \equiv \beta_m^*,$$

when the cyclic word S is reduced (E, β). That is to say the cyclic word $R(\alpha, \alpha')$ is simple.³⁰

Conversely, if R is a simple word it is obvious that γ is a free group, and we have the theorem:

THEOREM 4. *The group generated by $\alpha_1, \dots, \alpha_m$, subject to a single relation*

$$R(\alpha, \alpha') = 1,$$

which is not vacuous, is a free group if, and only if, the cyclic word $R(\alpha, \alpha')$ is simple.

²⁸ Cf. F. Levi, *Math. Zeit.*, 37 (1933), 90-97.

²⁹ W. Magnus, *Journal für die r.u.a. Math.* 163 (1930), 142.

See also K. Reidemeister, *Einführung in die kombinatorische Topologie*, Brunswick (1932), p. 92.

³⁰ W. G., §1.

It follows from this theorem that one can tell whether or no such a group is a free group by the process of reduction (T).

5. A problem which remains open is to determine whether or no Γ is equivalent to Γ^* under the automorphisms of G , where Γ and Γ^* are sub-groups of the latter, generated by given sets of elements (α) and (α^*). If Γ and Γ^* , generated by finite sets (α) and (α^*), are equivalent, one might hope to identify them by sequences of reductions (N) and (T), applied to (α) and (α^*), followed by level transformations. The following example shows that one cannot always do this

Let

$$\alpha_1 = (ab^{-1})^2 b^{-2}(ab^{-1})^2 a^3, \quad \beta_1 = a^{-3}b^{-5},$$

and

$$\alpha_2 = a^2b^{-2}a^2b^{-5}, \quad \beta_2 = (ab)^{-3} b^{-5}.$$

Then (α_1, β_1) and (α_2, β_2) are both unassailable by reductions of either kind. Since $L(\alpha_1, \beta_1) = 21$ and $L(\alpha_2, \beta_2) = 22$ they are not interchangeable by level transformations. All the same, if

$$\alpha = a^2b^{-2}a^2(ab)^3, \quad \beta = (ab)^{-3} b^{-5},$$

$(\alpha_1, \beta_1) \rightarrow (\alpha, \beta)$ in the automorphism $(a \rightarrow ab, b \rightarrow b)$, while $(\alpha_2, \beta_2) = (\alpha\beta, \beta)$.

BALLIOL COLLEGE,
OXFORD